

SLoMO Reproducibility Documentation

Abstract. This report releases a movie-level pipeline for open-ended long-video question answering. Each film supplies a timestamped transcript, chronological visual observations, and all of its questions in one request, yielding a validated JSON answer bank. The release includes code, chart data, tests, and implementation notes. Data, weights, predictions, credentials, and evaluation records remain external inputs.

Keywords: long-video QA · movie-level inference · reproducibility

1 System Overview

The movie is the context unit. A shared request preserves aliases, relationships, event order, causes, and ending state across questions. The release provides inference and submission utilities, tests, configuration guidance, and evidence-bound result plots. Private inputs remain external.

2 Method

For movie m , shared evidence E_m and its complete question set Q_m are mapped to an answer bank $\{a_{m,1}, \dots, a_{m,n_m}\}$. The runner groups rows by `video_id`, parses timestamped subtitle cues, samples 16 chronological frames, and sends all movie questions in one request. The model returns a JSON object keyed by the original IDs. The prompt requests concise, evidence-supported English answers and cross-question consistency. Frames are JPEG encoded at a fixed maximum side and marked with timestamps. Long transcripts use a head-and-tail bound.

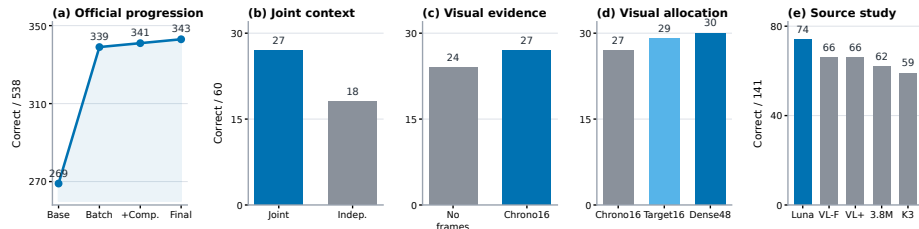


Figure 1. Evidence dashboard. (a) Historical official score progression. (b) Joint movie context versus independent question calls. (c–d) Visual configurations around the shared movie context. (e) Frozen source comparison. Values are correct answers with the panel denominator shown on the ordinate.

3 Inference Pipeline

The pipeline groups annotations, parses timestamped subtitles, samples the chronological frame pack, sends one movie-level request, and validates the complete JSON bank before persistence. Transient provider failures may be retried. Each persisted movie bank has complete question-ID coverage and non-empty answers.

4 Reproducibility

The runner accepts an OpenAI-compatible Responses endpoint and reads its key only from a process environment variable. The conversion utility writes the required `question_id,prediction` CSV. The audit utility checks the header, exact ID coverage, uniqueness, and non-empty predictions. Public and private splits are supplied independently.

Component	Configuration
Context	One movie, all questions jointly
Evidence	Timestamped transcript + 16 chronological JPEG frames
Frame encoding	Maximum side 512 px, JPEG quality 82
Generation	Temperature 0, high reasoning effort
Output	JSON bank, then <code>question_id,prediction</code> CSV

5 Results and Ablations

Figure 1 organizes the released evidence into five compact studies. Joint movie answering reaches 27/60, compared with 18/60 for independent calls, a gain of nine correct answers. The same visual budget with chronological frames reaches 27/60, and targeted and dense frame configurations reach 29/60 and 30/60. The source study anchors the chosen movie-batched answerer at 74/141. Together, these results motivate a coherent movie-level answer bank as the central inference unit.

Operating scope. The implementation targets movie-level QA with timestamped text, chronological visual observations, and an OpenAI-compatible model endpoint. Public sources are the SLoMO, EvalAI, and SF20K pages.